

Pre-Mid-term
Linear Algebra (MTH100), Section B
IIT Delhi, Monsoon 2024
September 20, 2024

Maximum marks: 50

Maximum Duration: $1\frac{1}{4}$ hours

Instructions

- (i) Attempt all five questions. All questions carry equal marks (parts may have different marks).
- (ii) All your intermediate steps and calculations must be clearly mentioned.
- (iii) Marks for proof based questions will depend on the logical progression of the steps. You may quote any propositions, or theorems covered in the lectures and tutorials. You are not allowed to use determinant or any other concepts which have not been introduced in the lectures of Section B. However, if you use any other result from outside the lectures, you have to prove that result, otherwise, it will not be considered.
- (iv) Write clearly and identify each part of your answers for the benefit of the tutors.

Notations: $I_{n \times n}$ is the identity matrix of size $n \times n$.

1. Consider the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ and $\mathbf{b} \in \mathbb{R}^4$ be any vector.

- (a) Find a unit lower triangular matrix L and an upper triangular matrix U such that $A = LU$. (6 points)
 - (b) Using part (a) decide when the system $A\mathbf{x} = \mathbf{b}$ is consistent. (2 points)
 - (c) Determine the set of all solutions of this system whenever it is consistent. (2 points)
2. In each of the following problems, provide a reason from the propositions or theorems done in class supporting your answer.
- (a) Can a square matrix with two identical rows be invertible? (2.5 points)

$b_{11} \ b_{12} \ b_{13} \ b_{14} \ b_{15}$
 $b_{21} \ b_{22} \ b_{23} \ b_{24} \ b_{25}$

$\begin{bmatrix} A & I & B^{-1} \end{bmatrix}$

- (b) If $A \in M_{5 \times 5}(\mathbb{R})$ and the system $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^5$, is it possible that for some $\mathbf{c} \in \mathbb{R}^5$, the system $A\mathbf{x} = \mathbf{c}$ has more than one solution? (2.5 points)

- (X) Assume that $A \in M_{5 \times 7}(\mathbb{R})$ and $B \in M_{7 \times 5}(\mathbb{R})$ satisfy the equation $BA = I_{7 \times 7}$. Does the system $A\mathbf{x} = \mathbf{y}$ has 5 solutions whenever it is consistent? (2.5 points)

- (X) Let $A \in M_{3 \times 5}(\mathbb{R})$ with column vectors $\mathbf{A}_1, \dots, \mathbf{A}_5 \in \mathbb{R}^3$ satisfy the equation (2.5 points)

$$\mathbf{A}_1 + c_1 \mathbf{A}_2 + c_2 \mathbf{A}_3 + c_3 \mathbf{A}_4 + c_4 \mathbf{A}_5 = \mathbf{0},$$

for some $c_1, c_2, c_3, c_4 \in \mathbb{R}$. Can the rank of A be 5?

$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \cdot \mathbf{x} = \mathbf{y}$
 $\mathbf{x} = \mathbf{by}$

3. Assume all the matrices are $n \times n$. Decide TRUE OR FALSE? Whenever you claim that a statement is TRUE, you have to provide a proof. And, to disprove a statement (or equivalently, to say a statement is FALSE) you need to produce a counter-example.

- (a) If $AB = CB$ and $B \neq 0$, then $A = C$. (3 points)
 (b) $(A+B)(A-B) = A^2 - B^2$ if and only if $AB = BA$. (3 points)
 (c) There are no unit lower triangular matrix $L \in M_{2 \times 2}(\mathbb{R})$ and upper triangular matrix $U \in M_{2 \times 2}(\mathbb{R})$ such that (4 points)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = LU.$$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

4. (a) If A is an $m \times n$ matrix over $\mathbb{R}$ and $A\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n$, show that $A = 0$ (here, 0 denotes the zero matrix - the matrix whose every entry is zero). (5 points)
 (b) If A and B are $m \times n$ matrices over $\mathbb{R}$ and $A\mathbf{x} = B\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, show that $A = B$. (5 points)
 5. (a) Suppose that A is a square matrix of size 10×10 satisfying the equation (5 points)

$$A^3 - 3A + I_{10 \times 10} = 0.$$

Show that A is invertible and find A^{-1} .

- (b) Give a matrix that has a left inverse but no right inverse. (2.5 points)
 (c) Give a matrix that has right inverse but no left inverse. (2.5 points)